Maastricht treaty

The Maastricht Treaty formally establishes the path to the euro and the EMU

Three important leaders led to the sign of TM: Mitterand, Delors, Khol

Mitterand: President of France

Khol: Chancellor of Germany

The two leaders mediated the different goals and views across the states

Delors: President of the European Commission. "Delors Report" (1989) is a landmark and the blueprint for the Maastricht Treaty

The Treaty was signed in Maastricht on 7 February 1992 and enforced 1 Nov 1993.

The provisions relating to EMU (based on Delors Report) constitute the largest part of the TM (phases, convergence, ECB, ECB independence, fiscal coordination, etc.).

Other provisions of TM:

- Introduction of the principle of subsidiarity;
- Revision of the Structural Funds for the Cohesion Objective;
- Extension of the Common Market to transport, health, culture,…

Two principles: graduality and convergence

The TM provides for the establishment of a single currency in three successive phases:

the first stage -liberalising capital movements- begins on 1 July 1990;

the second stage begins on 1 January 1994 with the convergence of the economic policies of the Member States;

the third stage begins on 1 January 1999 with the creation of a single currency and the establishment of a European Central Bank (ECB).

The TM provides that entry into the UM is subject to the fulfilment of the so-called convergence criteria.

The rationalization of the Maastricht convergence requirements

- The Treaty stresses *macro*-economic convergence but only related to monetary policy and public finance
  - inflation
  - interest rates
  - budgetary policies
- An important worry is inflation and conditions that may lead to inflationary pressure

Convergence criteria

For each candidate country:

(1) inflation rate $\leq$ average of three lowest inflation rates in the group of candidate countries + 1.5%;

(2) long-term interest rate $\leq$ average observed in the three low-inflation countries + 2%;

(3) joined the exchange rate mechanism of the EMS and did not experience a devaluation during the two years preceding the entrance into EMU;

(4) government budget deficit $\leq$ 3% of its GDP

if this condition is not satisfied:

budget deficit should be declining continuously and substantially and come close to the 3%,

or the deviation from the reference value (3%) 'should be exceptional and temporary and remain close to the reference value', art. 104c(a));

(5) government debt $\leq$ 60%of GDP

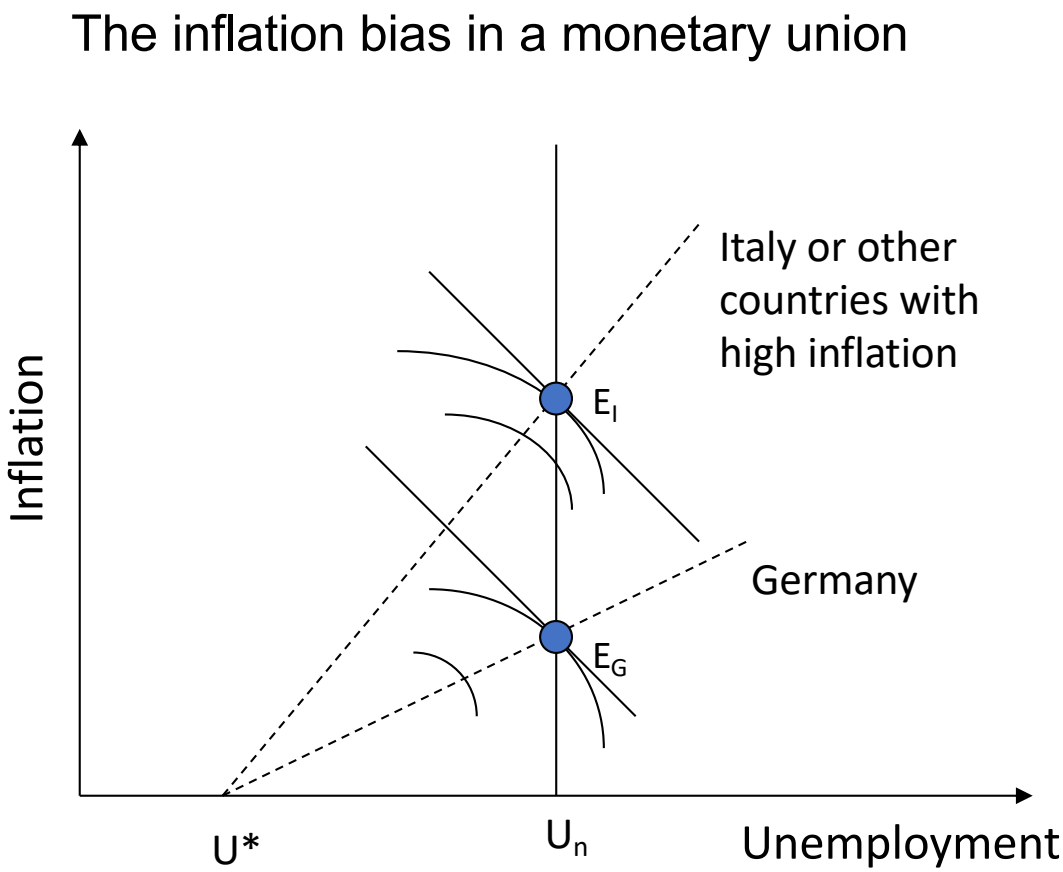
if this condition is not satisfied:

government debt should 'diminish sufficiently and approach the reference value (60%) at a satisfactory pace', art. 104c(b)).

1. Inflation convergence

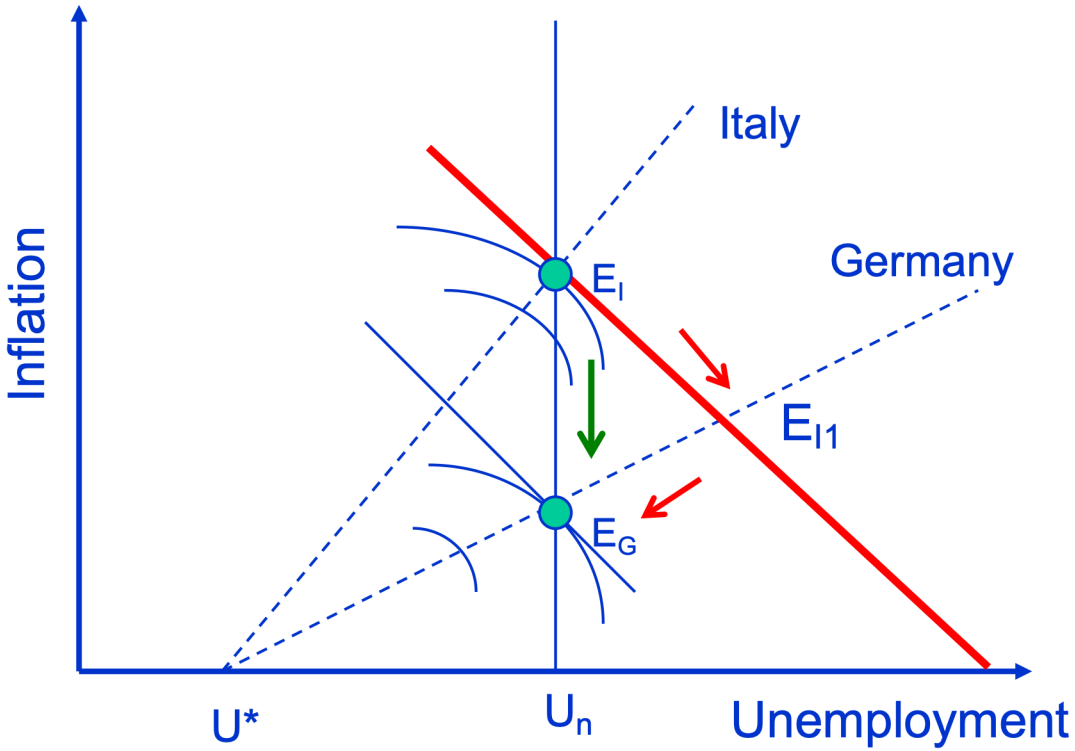
- Avoid that the future monetary union could have an inflationary bias.
- The analysis is embedded in the Barro - Gordon model.

The inflation bias in a monetary union



- Before EMU Germany has low inflation; Italy has high inflation.
- Difference is due to different preferences concerning inflation and unemployment.
- Once in the Union, Germany and Italy will decide together.
- Inflation will reflect average preferences and will be located between E_I and E_G.
- Germany loses welfare; Italy gains welfare.
- Germany wants evidence from Italy that it has the same strong preference for price stability.
- Germany also wants ECB to be clone of Bundesbank

The problem of reducing inflation



- The reduction of inflation for Italy and the way the country can move from E_I to E_G is a relevant issue
- Ideally: reduction in inflation and no rise in unemployment (green arrow)
- In practice: reduction along the Phillips curve: red arrow to E_{I1} and then E_G.
- How log will this process take?
- Will it determine a rise of the natural rate of unemployment?

2. Exchange rate convergence (no-devaluation requirement)

- It prevents countries from manipulating their exchange rates
  - e.g. so as to have more favorable (depreciated) exchange rate in the Union
- Note:
  - According to the Treaty, countries should maintain their exchange rates within the 'normal' band of fluctuation and without changing that band, which implies no change of the central rate, during the two years preceding their entry into the EMU.
  - Since August 1993, the 'normal' band within the EMS was -/+15%.

3. Interest rate convergence

- Excessively large differences in the interest rates prior to entry could lead to large capital gains and losses at the moment of entry into EMU .
- However, these gains and losses are likely to occur prior to entry because the market will automatically lead to a convergence of long-term interest rates as soon as the political decision is made to allow entry of the candidate member country.
- Indeed, given the inflation criteria, the no devaluation criteria and free capital movements, interest rates were going to converge (recall the interest rate parity)

4-5. Budgetary convergence

- Deficit and debt criteria can be rationalized in a similar way
  - A country with a high debt-to-GDP ratio has an incentive to create surprise inflation.
  - The low debt country stands to lose and will insist that the debt-to-GDP ratio of the highly indebted country be reduced prior to entry into the monetary union.
  - The high debt country must also reduce its government budget deficit.

Government budget and debt

Let:

B = nominal debt

D = nominal deficit = expenses - revenues

Y = nominal GDP.

$$B_t = B_{t-1} + D_t$$

Let us distinguish between the Government primary expenditure and the expenditure due to interest payment on the existing debt

$$D_t = G_t - T_t = \left(G_t^P + i_t B_{t-1}\right) - T_t = i_t B_{t-1} + \left(G_t^P - T_t\right) = i_t B_{t-1} + D_t^P$$

Where i_t is the average interest rate on the existing debt

$$B_t = B_{t-1} + i_t B_{t-1} + D_t^P$$

$$B_t = B_{t-1}\left(1 + i_t\right) + D_t^P$$

$$also:$$

$$B_t - B_{t-1} = 0 \quad if \quad i_t B_{t-1} = -D_t^P = A^P$$

Nominal debt remains constant if the primary surplus is sufficient to cover interests payment.

What about the debt in real terms?

The real value of debt

$$\frac{B_t}{P_t} = \frac{B_{t-1}(1+i)}{P_t} + \frac{D_t^P}{P_t}$$

$$\frac{B_t}{P_t} = \frac{B_{t-1}}{P_{t-1}} \frac{(1+i)}{\frac{P_t}{P_{t-1}}} + \frac{D_t^P}{P_t}$$

$$\frac{B_t}{P_t} = \frac{B_{t-1}}{P_{t-1}} \frac{(1+i)}{\frac{P_t}{P_{t-1}} - 1 + 1} + \frac{D_t^P}{P_t}$$

$$\widetilde{B}_t = \widetilde{B}_{t-1} \frac{1+i_t}{1+\dot{p}_t} + \widetilde{D}_t^P$$

NOTE: A rise in inflation reduces the real value of debt, *ceteris paribus*

Gov may be temptated to create unexpected inflation.

Debt monetization is prevented by the EU Treaty

- In addition, countries with a large debt face a higher default risk.
- Once in the union, this will increase the pressure for a bailout in the event of a default crisis.
- *No-bailout* clause was incorporated into the Maastricht Treaty.
- But is this clause credible?

Maastricht criteria on public finance

$$\frac{B_t}{Y_t} = \frac{B_{t-1}}{Y_t} + \frac{D_t}{Y_t}$$

$$\frac{B_t}{Y_t} = \frac{B_{t-1}}{Y_{t-1}} \frac{1}{\frac{Y_t}{Y_{t-1}}} + \frac{D_t}{Y_t}$$

$$\frac{B_t}{Y_t} = \frac{B_{t-1}}{Y_{t-1}} \frac{1}{\frac{Y_t}{Y_{t-1}} - 1 + 1} + \frac{D_t}{Y_t}$$

$$(1) \quad b_t = b_{t-1} \frac{1}{1 + g_{N,t}} + d_t$$

No distinction between primary expenditure and interest expenditure

Debt and deficit are considered relative to GDP (Y)

Small letters indicate GDP ratios and g_N is the nominal GDP rate of growth.

Notice: equation (1) is a first order difference equation

The solution of the difference equation is characterized by an equilibrium solution and a deviation from the equilibrium.

As time passes, the deviation can either die out or become bigger

The equilibrium value is such that: $b_t = b_{t-1} = \bar{b}$

Imposing this in equation (1) we get :

$$\bar{b} = \frac{\bar{b}}{1+g_N} + d$$

$$\bar{b} + \bar{b}g_N = \bar{b} + d + dg_N$$

$$\bar{b}g_N = d + dg_N \rightarrow \bar{b} = \left(\frac{1+g_N}{g_N}\right)d$$

where everything is assumed constant.

Since dg_N is a small number, for convenience we approximate the equilibrum solution to the following:

$$\bar{b}g_N = d \rightarrow g_N = \frac{d}{\bar{b}}$$

In equilibrium (when debt is stable) there exists a precise relationship between deficit and debt as a percentage to GDP and GDP growth.

What is the relationship implicitly assumed in the Masstricht Treaty?

Given $g_N = \frac{d}{b}$ and b=0.6 e d=0.03 → g_N=0.03/0.6 = 0.05

If nominal GDP grows at 5%, real GDP grows at

5%-equilibrium inflation rate → 5-2=3%

Thresholds of 3% and 60% are arbitrary

Numerical precision of budgetary requirements is difficult to rationalize

- 3% and 60% budgetary norms have been derived from formula determining budget deficit needed to stabilize government debt:

$$d = gb$$

b = (steady state) level at which the government debt is to be stabilized (in per cent of GDP),

g = growth rate of nominal GDP, and

d = government budget deficit (in per cent of GDP).

In order to stabilize the government debt at 60% of GDP the budget deficit must be brought to 3% of GDP if and only if the nominal growth rate of GDP is 5% (0.03 = 0.05 x 0.6).

Arbitrary nature of the rule

- The rule is quite arbitrary on two counts.
  - Why should the debt be stabilized at 60%?
    - The only reason was that at the time of the Maastricht Treaty negotiation this was the average debt-to-GDP ratio in the European Union.
  - The rule is conditioned on the future nominal growth rate of GDP
    - If the nominal growth of GDP increases above (declines below) 5%, the budget deficit that stabilizes the government debt at 60% increases above (declines below) 3%.

Should we worry about high public debts?

The dynamics of the debt/GDP ratio.

Start from the debt in real terms and divide by GDP in real terms.

$$\widetilde{B}_t = \widetilde{B}_{t-1}\frac{1+i_t}{1+\dot{p}_t} + \widetilde{D}_t^P$$

For convenience let: $\frac{1+i_t}{1+\dot{p}_t} = 1+r_t \cong 1+i-\dot{p}_t$

$$\widetilde{B}_t = \widetilde{B}_{t-1}(1+r_t) + \widetilde{D}_t^P$$

Divide by GDP in real terms:

$$b_t = b_{t-1}\frac{1+r_t}{1+g_t} + d_t^P$$

where g_t is the rate of growth of GDP in real terms

The equilibrium solution is:

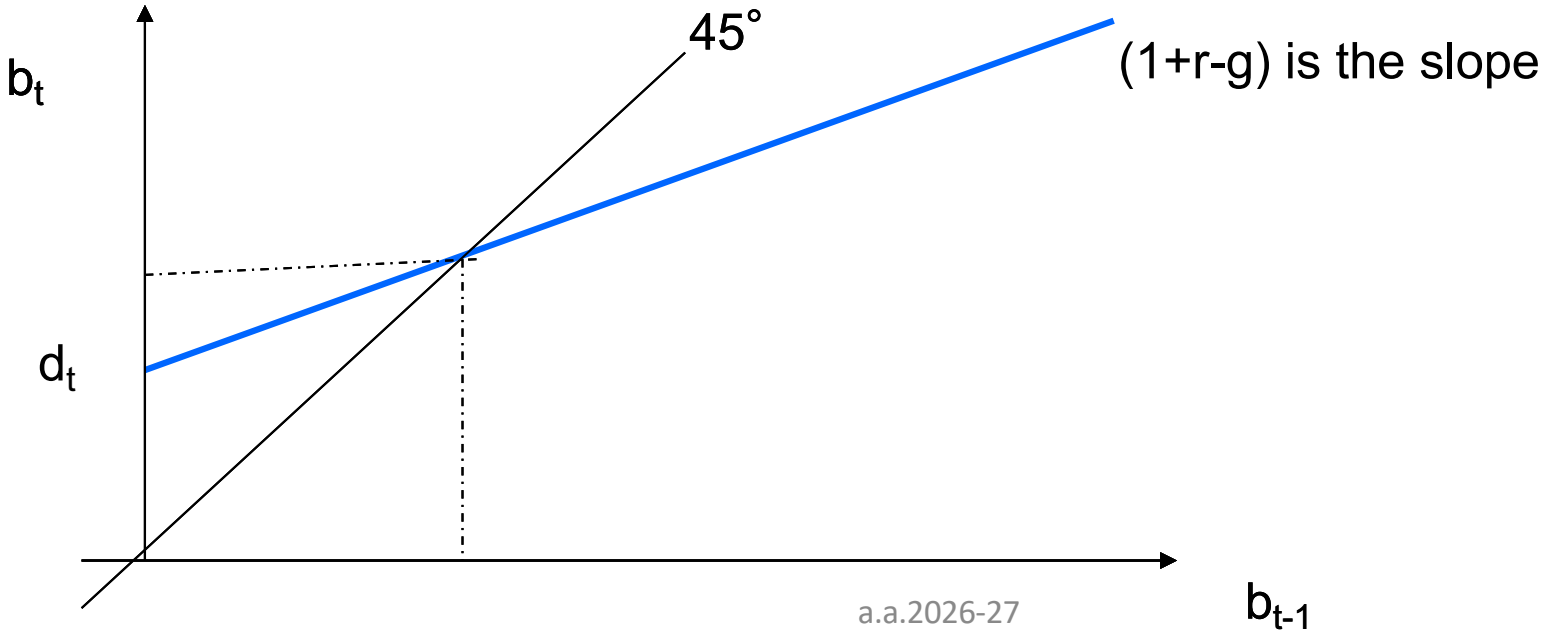
$$\bar{b} = \bar{b}\left(\frac{1+r}{1+g}\right) + d^P \rightarrow \bar{b}\left(1-\frac{1+r}{1+g}\right) = d \rightarrow \bar{b} = d\frac{1+g}{g-r}$$

Simple graphical analysis of the debt dynamics

For convenience let the slope coefficient be approximated by (1+r-g):

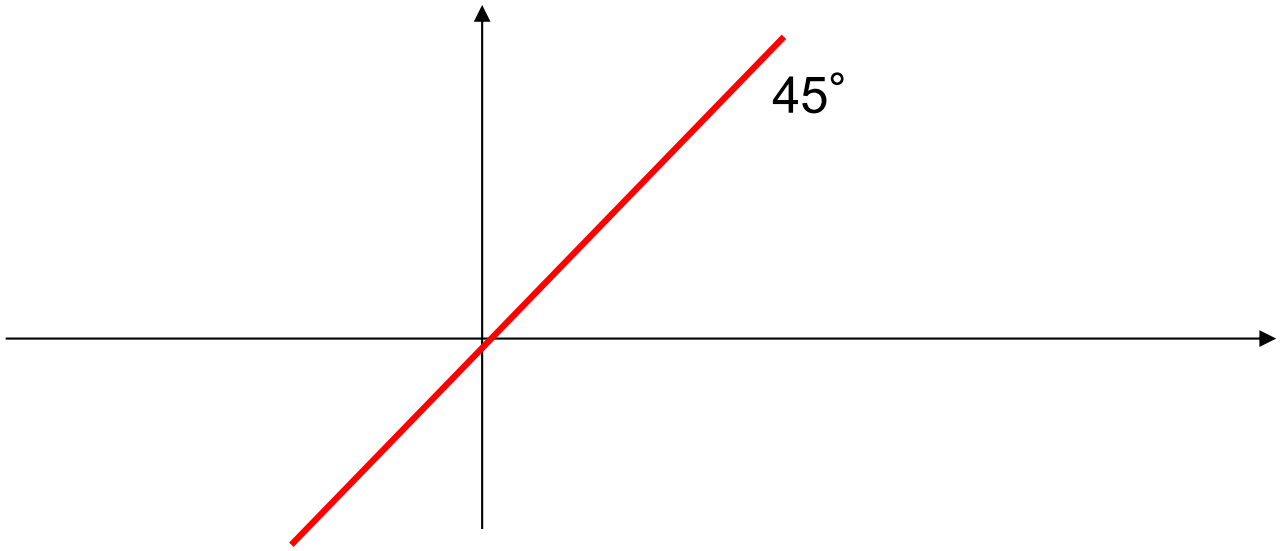
$$b_t = b_{t-1}\left(1 + r_t - g_t\right) + d_t^P$$

This difference equation can be represented as a line in the space: b_t, b_{t-1}



The locus of stationary equilibrium or stable debt/GDP

1. The 45° line represents equilibrium points

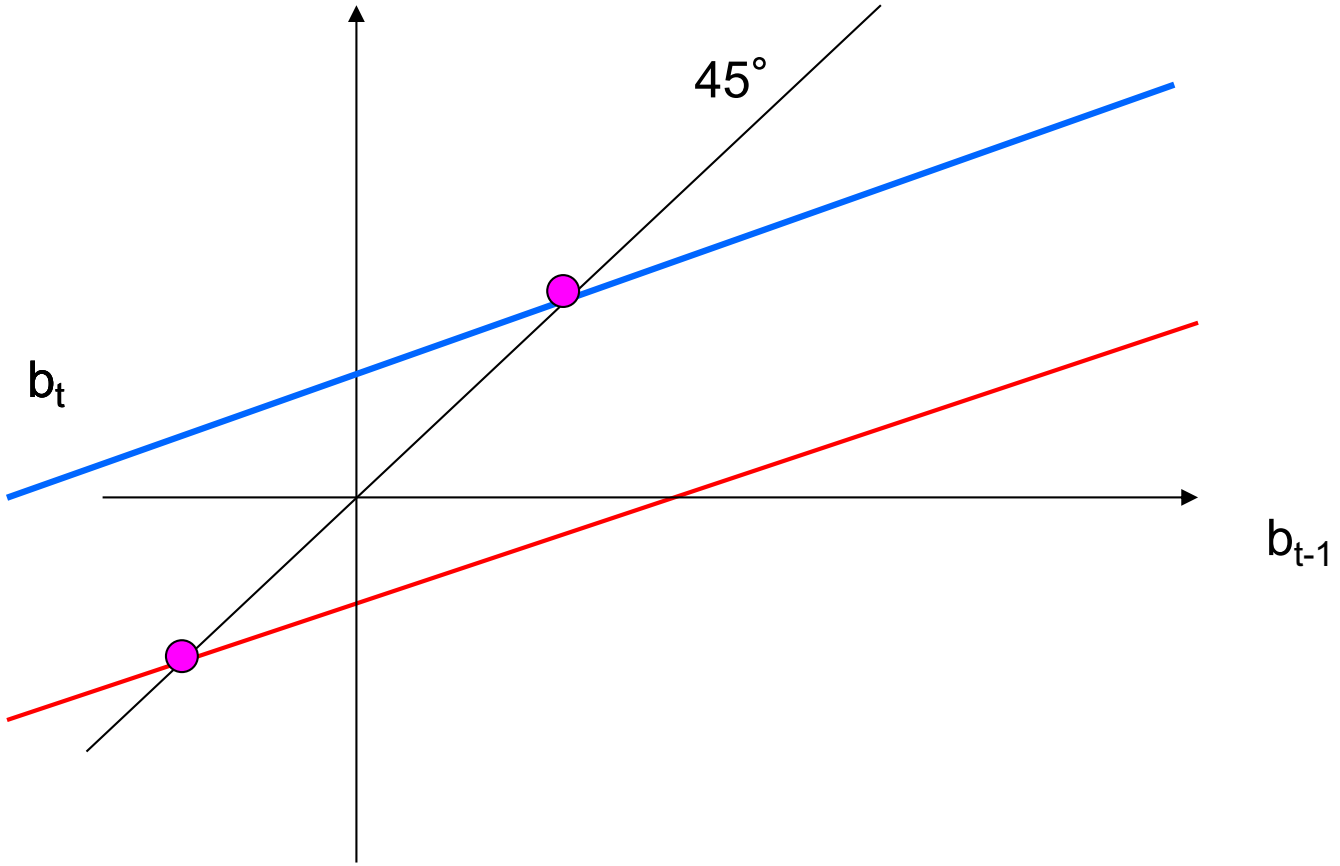


2.Debt is in equilibrium if :

$$\bar{b} = \bar{b}(1 + r - g) + d^P \rightarrow \bar{b} = \frac{d^P}{g - r}$$

Changes in primary deficit move the line – translation

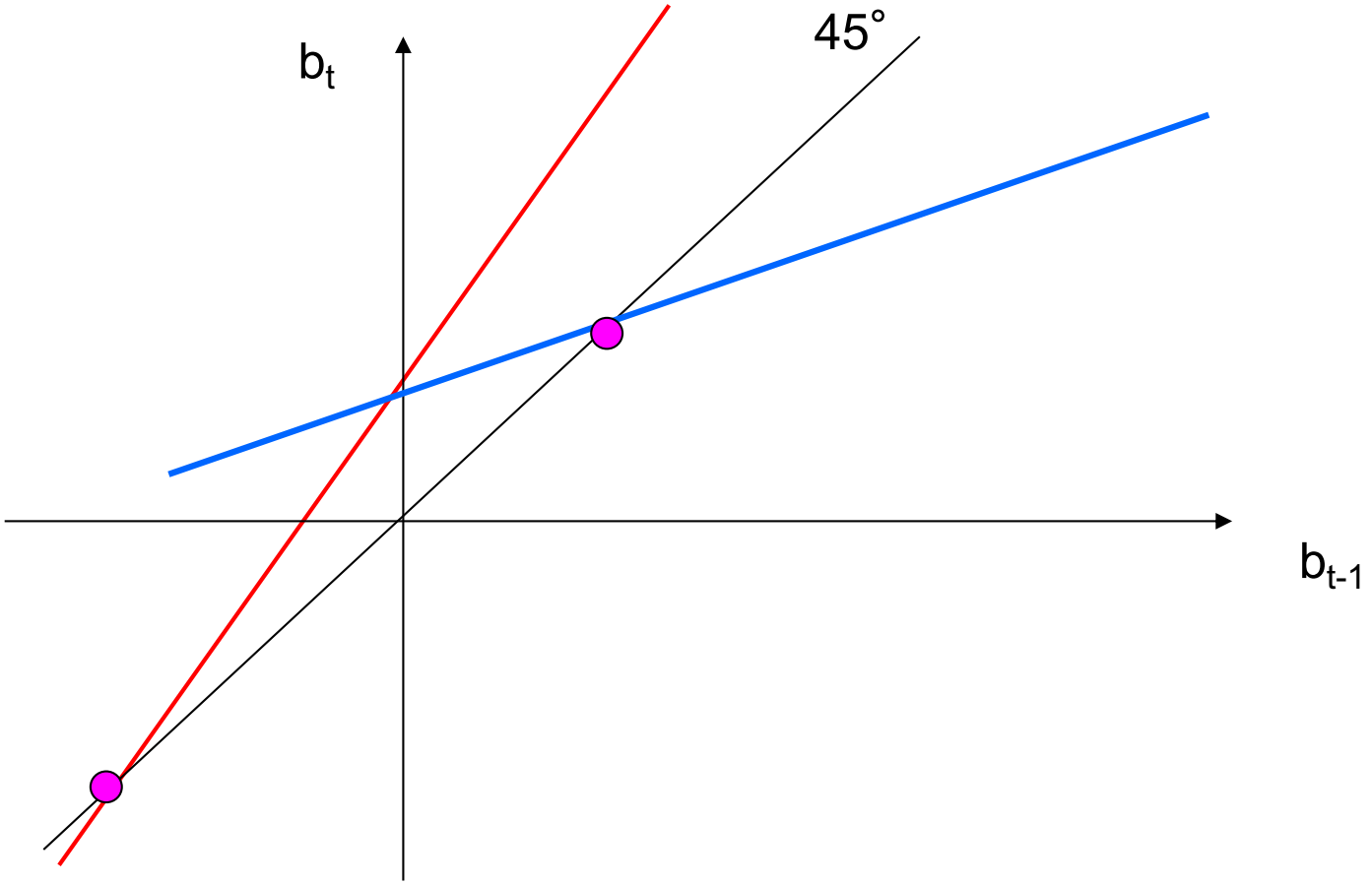
3. Intercept: postive if there is a **deficit primario/PIL,** negative is there is **a surplus**



b_{t-1}

4. **The slope of the debt line depends on the inequality between r and g.**

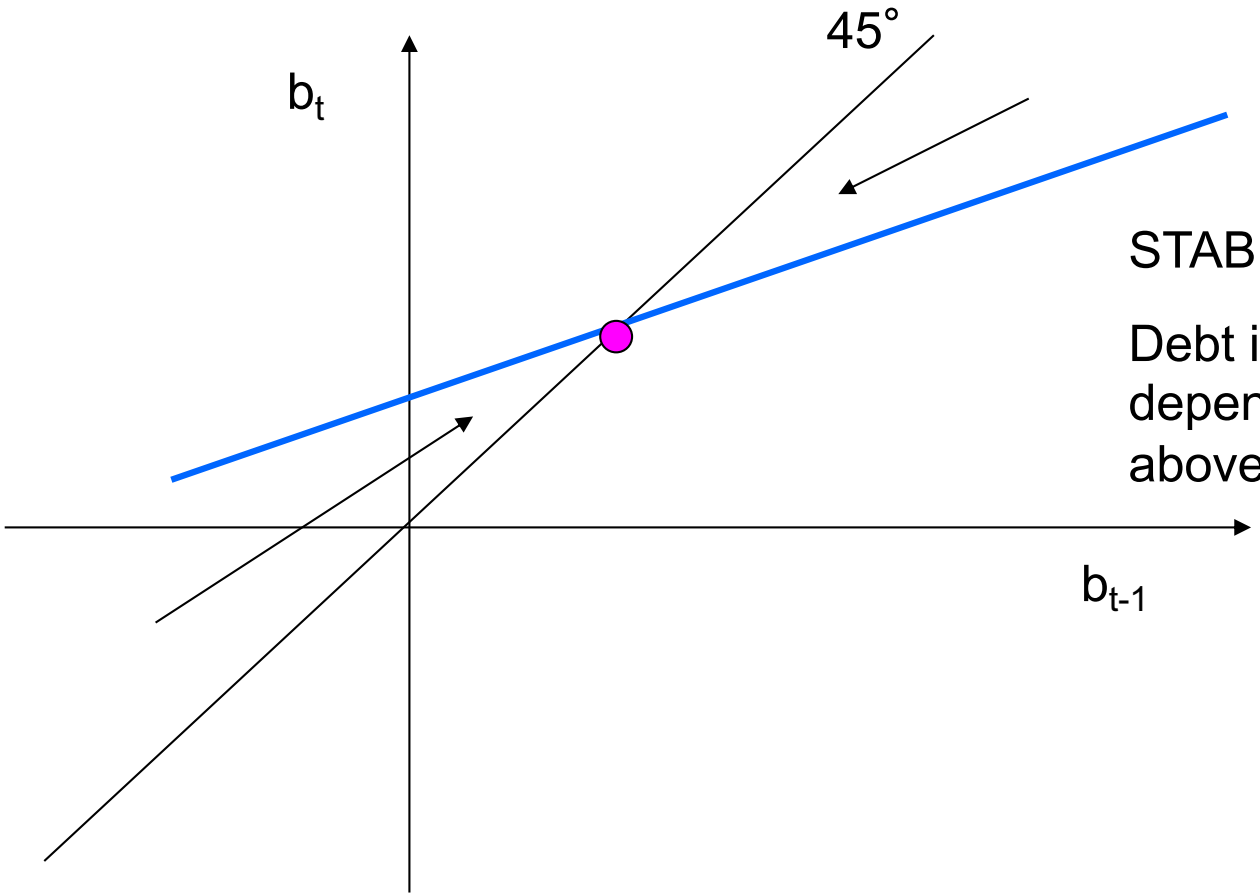
If g>r the slope is less than 1; if g<r slope is greater than 1



b_{t-1}

Debt dynamics (graphical analysis)

1° case: g>r and positive primary deficit

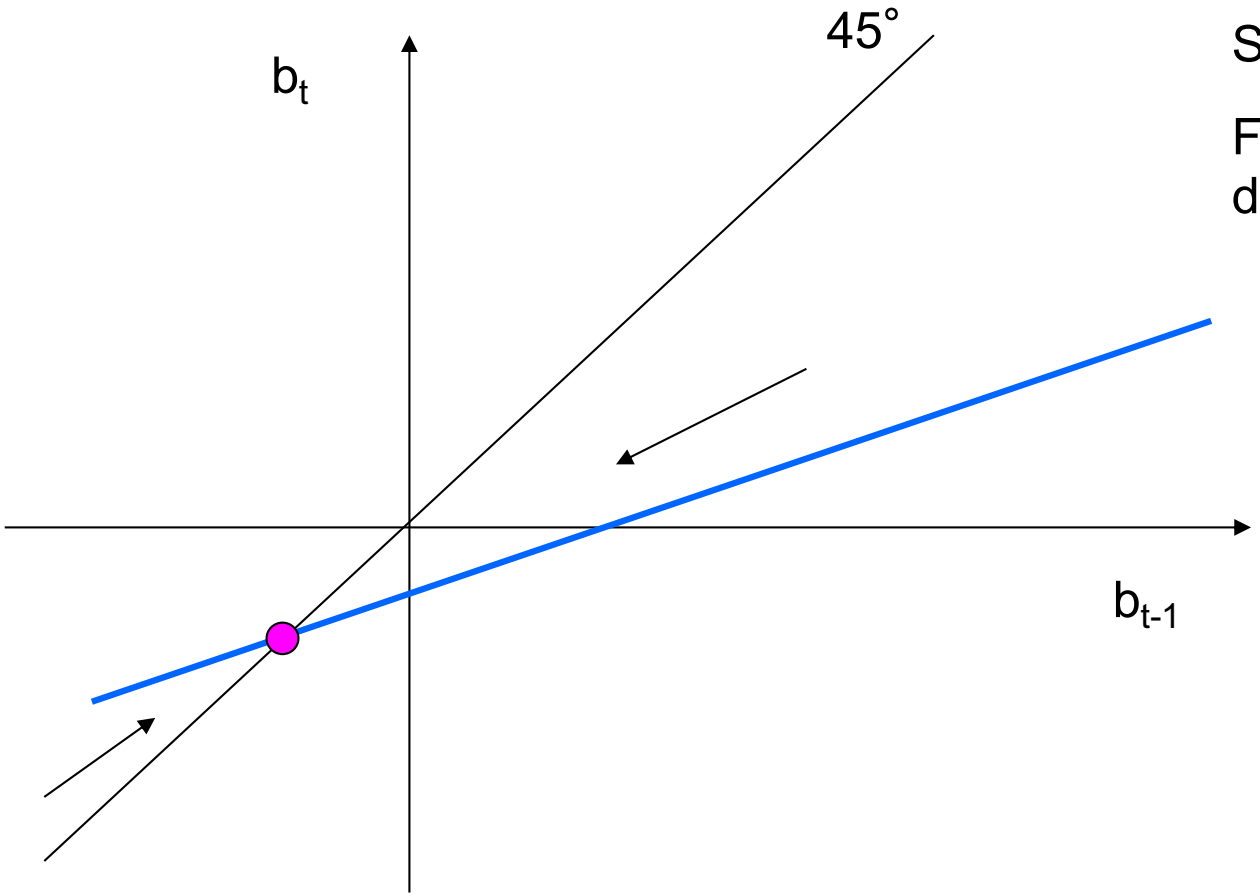


STABLE equilibrium

Debt increasing or decreasing depending on being below or above the equilibrium

Debt dynamics (graphical analysis)

2° case: g>r and negative primary deficit (surplus)

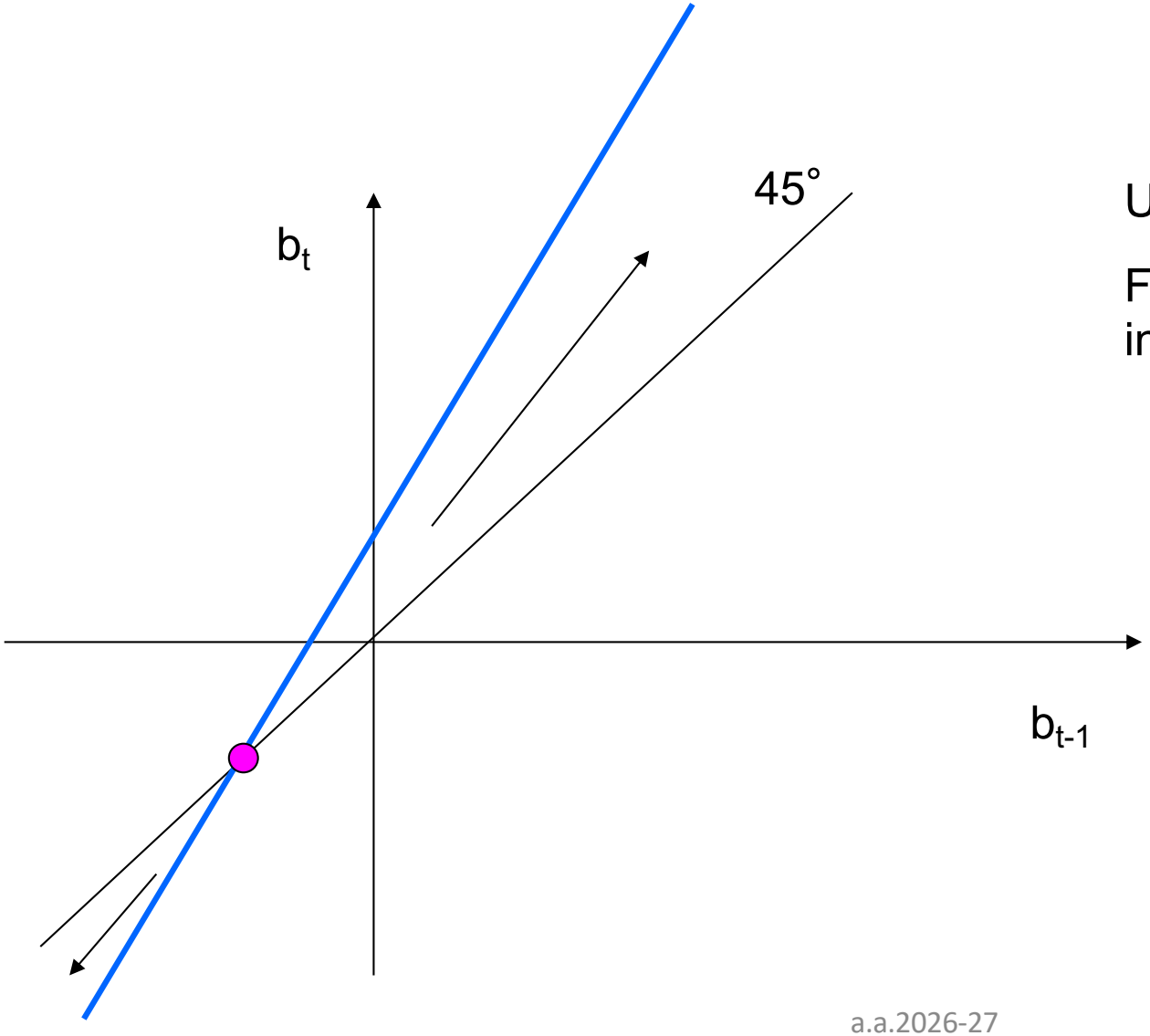


STABLE equilibrium

For all positive values, debt is decreasing

Debt dynamics (graphical analysis)

3° case: g<r and positive primary deficit

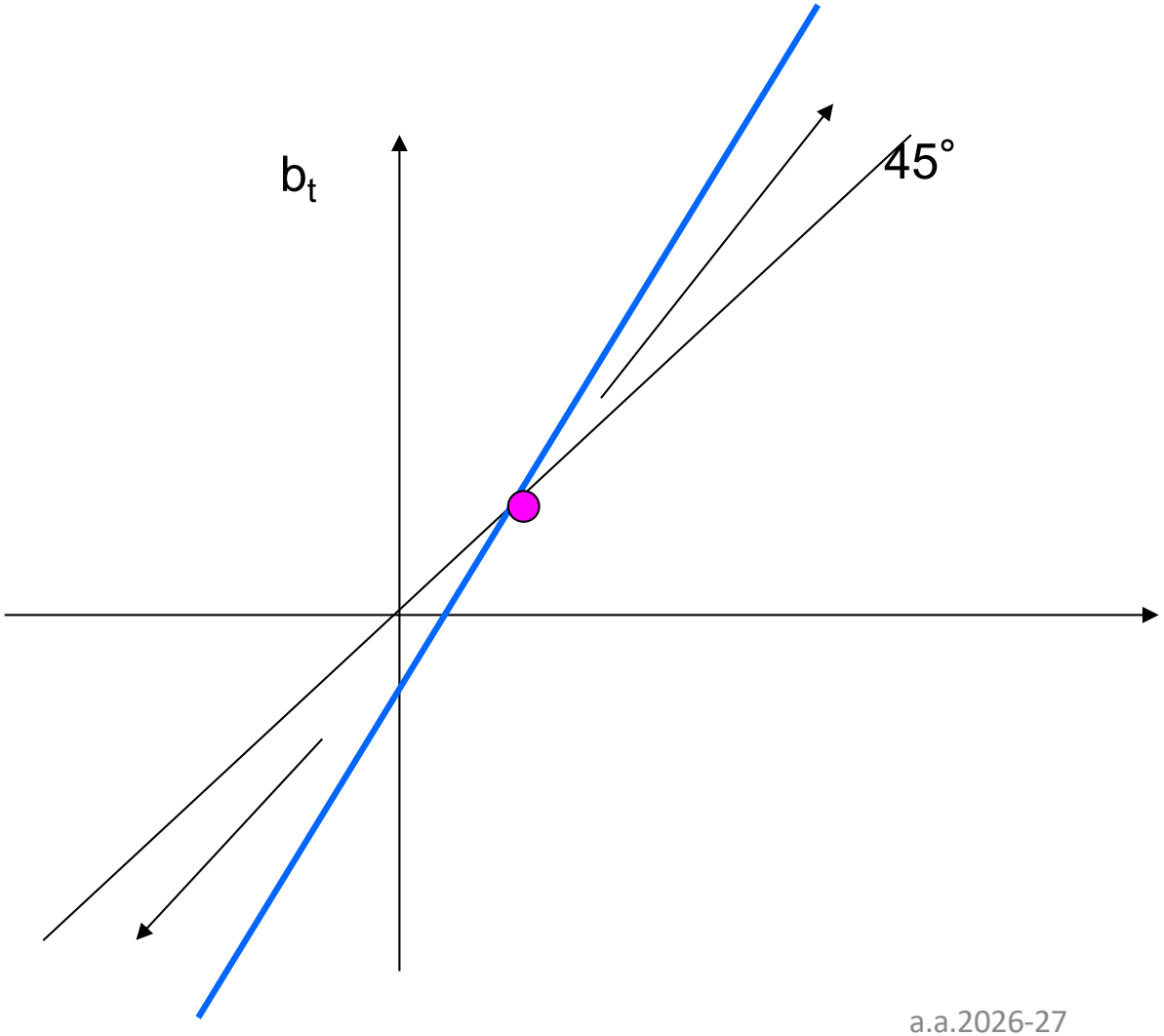


UNSTABLE equilibrium

For all positive values, debt is increasing

Debt dynamics (graphical analysis)

4° case: g<r and negative primary deficit (surplus)



UNSTABLE equilibrium

Debt is increasing (decreasing) if above (below) the equilibrium

How much effort does it take to stabilize the debt/GDP?

$$b_t = b_{t-1}\frac{1+r_t}{1+g_t} + d_t^P$$

$$\Delta b_t = b_t - b_{t-1} = b_{t-1}\left(\frac{1+r_t}{1+g_t} - 1\right) + d_t^P$$

$$(*)\ \Delta b_t = b_{t-1}\left(\frac{r_t - g_t}{1+g_t}\right) + d_t^P$$

Snowball effect

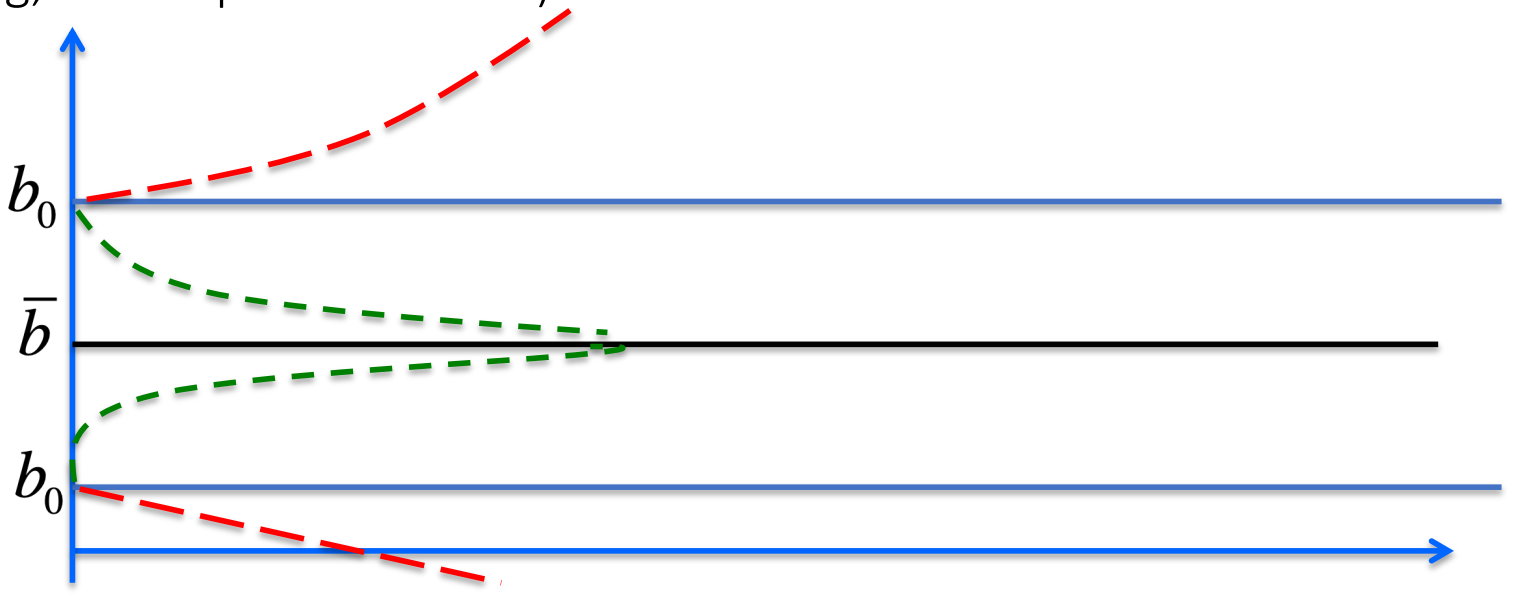
We can re-write (*):

$$(**)\ \Delta b_t = b_{t-1}\left(\frac{r_t - g_t}{1+g_t}\right) - a_t^P$$

In order for debt/GDP to decline or stop ($\Delta b <= 0$) primary surplus must be relatively high if r>g

Solution to the difference equation

(hp: g, r and dp are constant)



$$b_t = \left(b_0 - \bar{b}\right)\left(\frac{1+r}{1+g}\right)^t + \bar{b}$$

- Deviation of starting value (b_0) from the stationary state
- Evolution of the deviation
- Stationary state

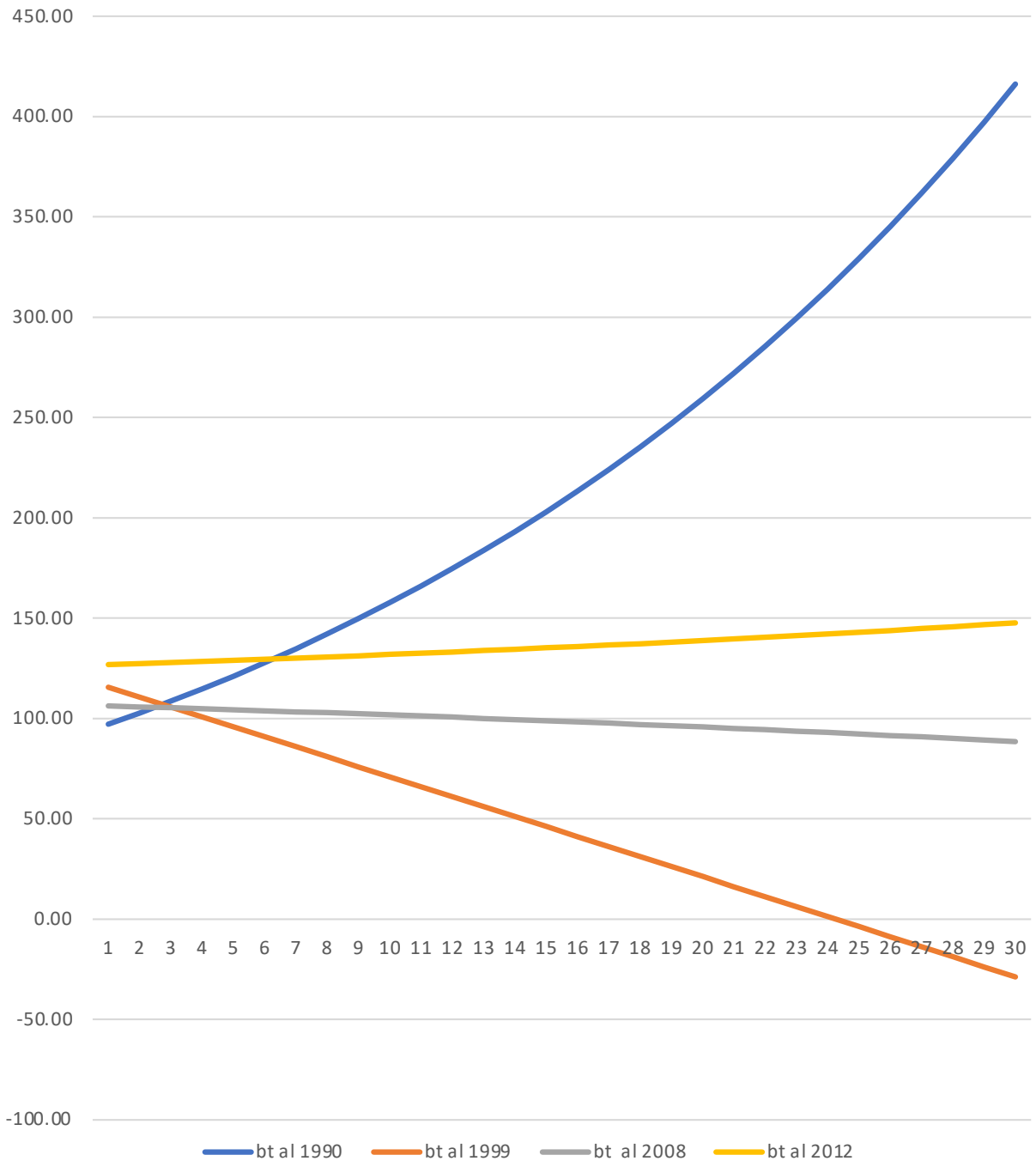
Public debt/GDP in Italy

	d primary	g	r	(1+r)/(1+g)	b stationary	b
2008	-2.5	-1.3	0.59	1.019	1.31	106.3
2009	0.8	-5.2	1.38	1.069	-0.12	116.1
2010	0.0	1.3	0.50	0.992	0.00	119.1
2011	-1.0	0.5	0.71	1.002	4.79	120.1
2012	-2.5	-2.5	-0.19	1.024	1.06	126.9
2013	-1.9	-1.9	0.88	1.028	0.67	129
2014	-1.6	-0.3	1.15	1.015	1.10	131.8
2015	-1.5	0.8	0.60	0.998	-7.56	131.5
2016	-1.5	0.9	0.65	0.998	-6.05	132

Source: Bollettino economico
Banca d'Italia
and Istat

- see excel file on debt dynamics

Debt/GDP dynamics forecasted on the basis of data as at 1990, 1999, 2008, 2012, on the assumption that g, r and primary deficit remain constant at their values as at the initial year.

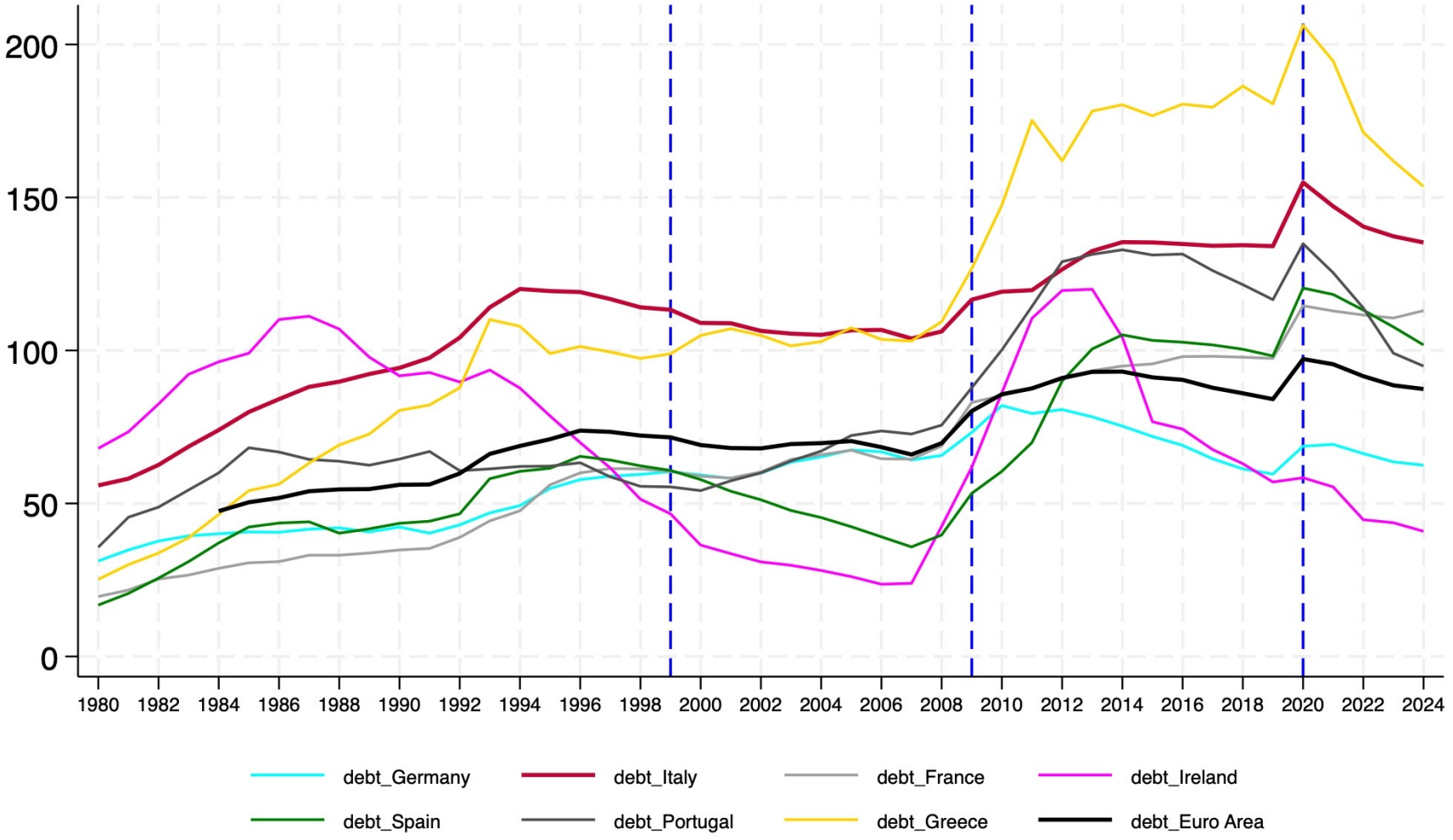


Government budget statistics

- https://www.bancaditalia.it/pubblicazioni/finanza-pubblica-ue/2022-finanza-pubblica-ue/statistiche_FPE_13122022.pdf

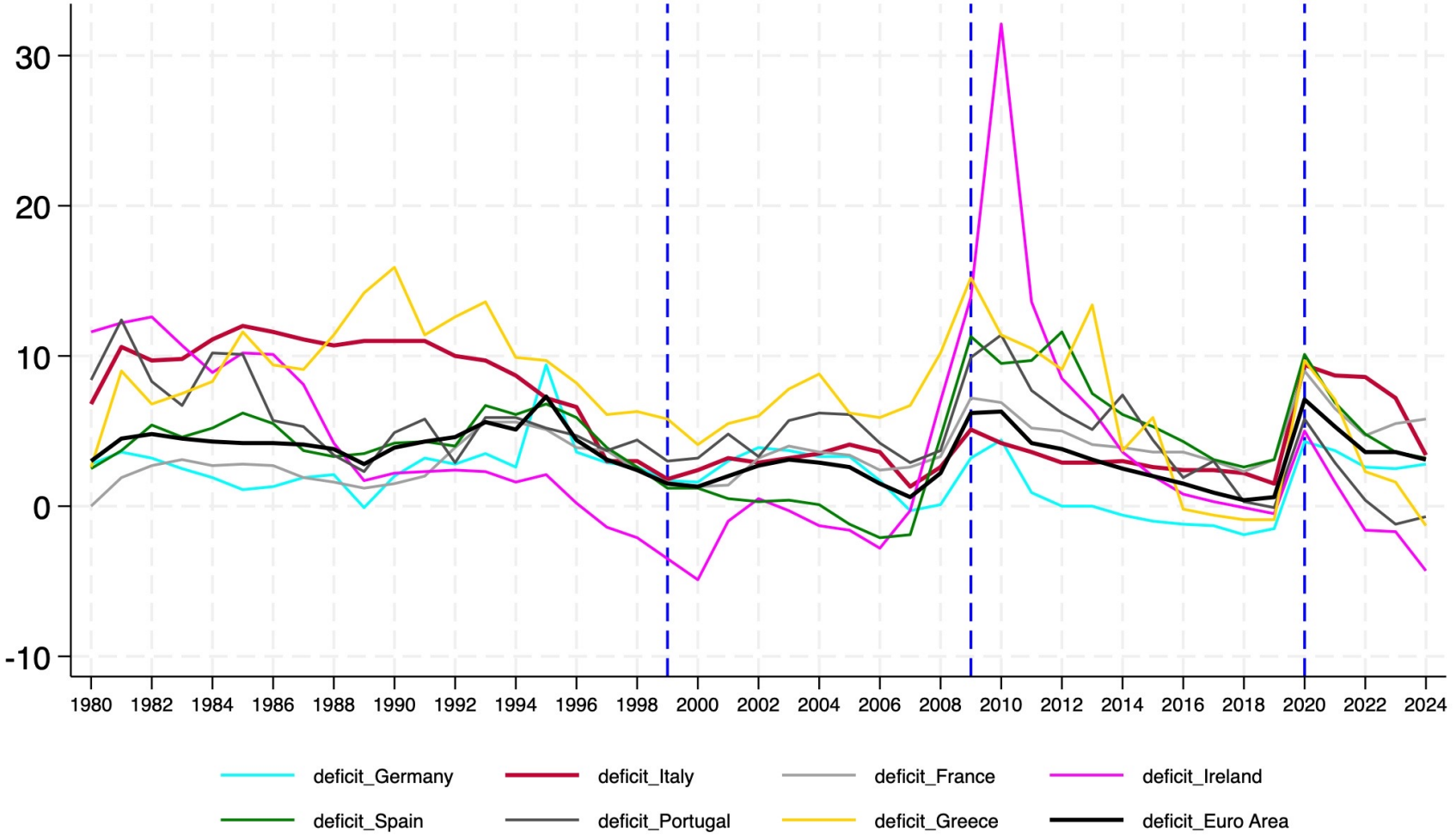
- A look at the evolution of public finances in the Eurozone

Public debt/GDP



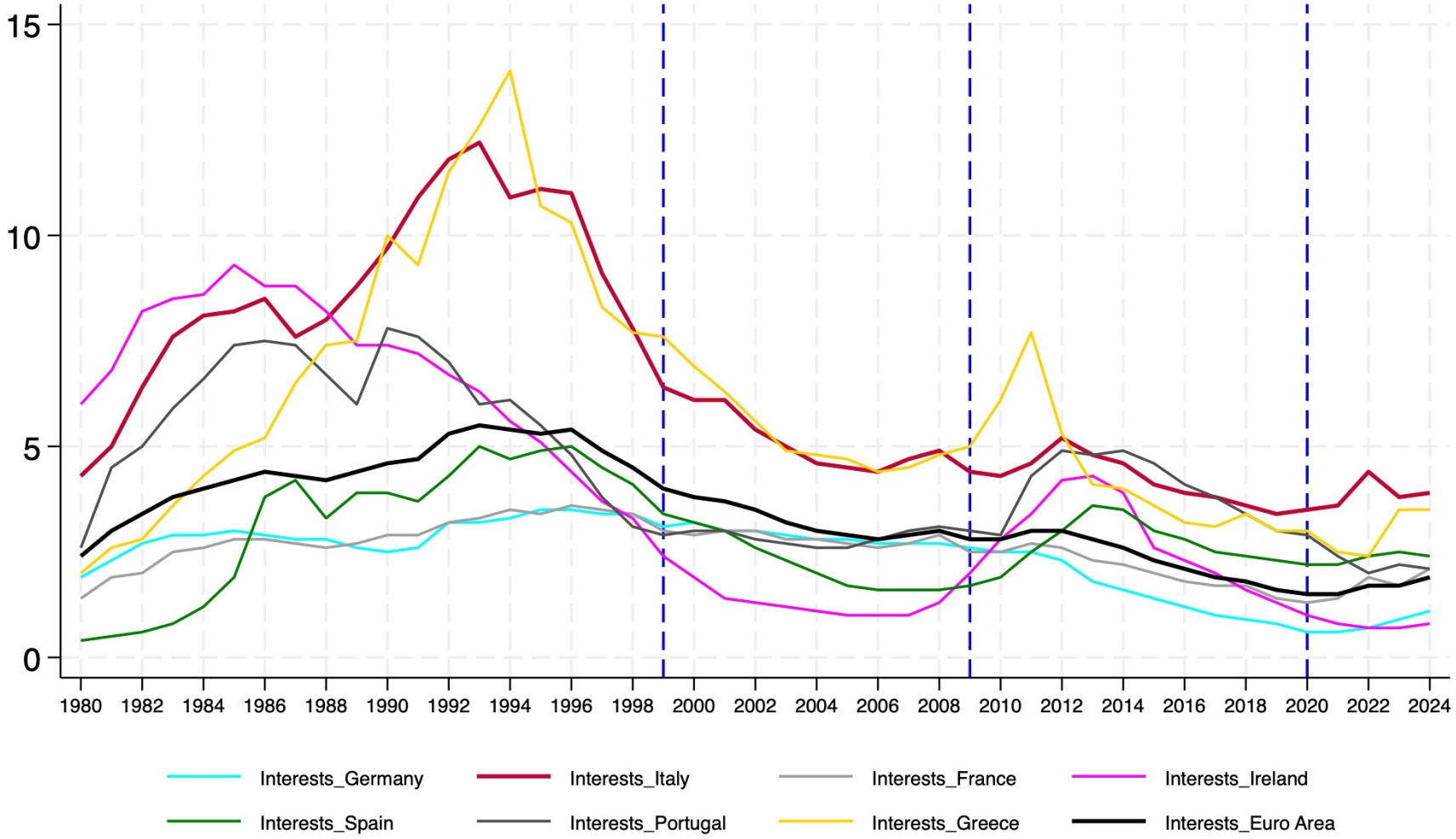
Source: Own elaboration on BdI data

Deficit/GDP



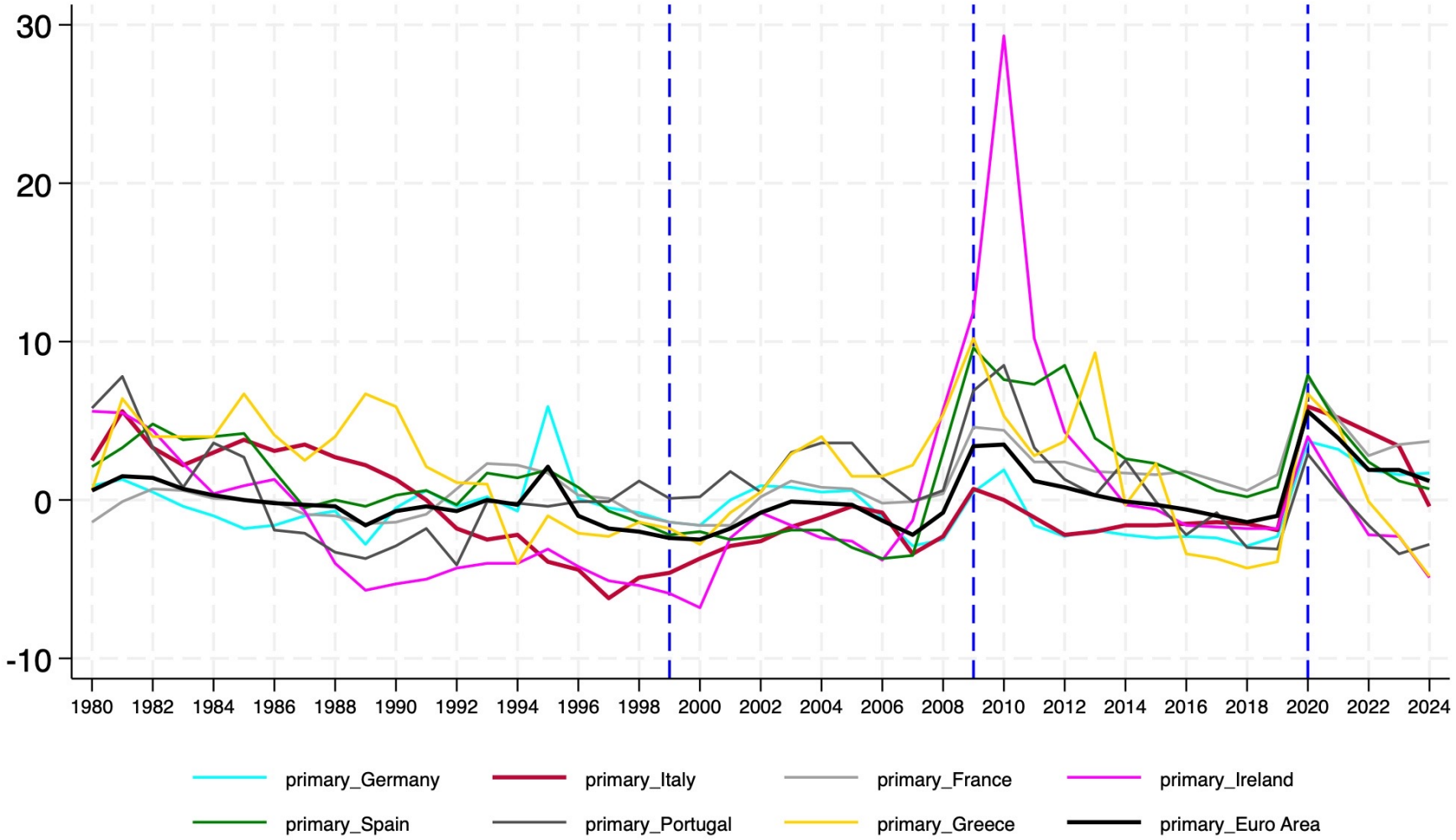
Source: Own elaboration on BdI data

Interests on public debt/GDP



Source: Own elaboration on BdI data

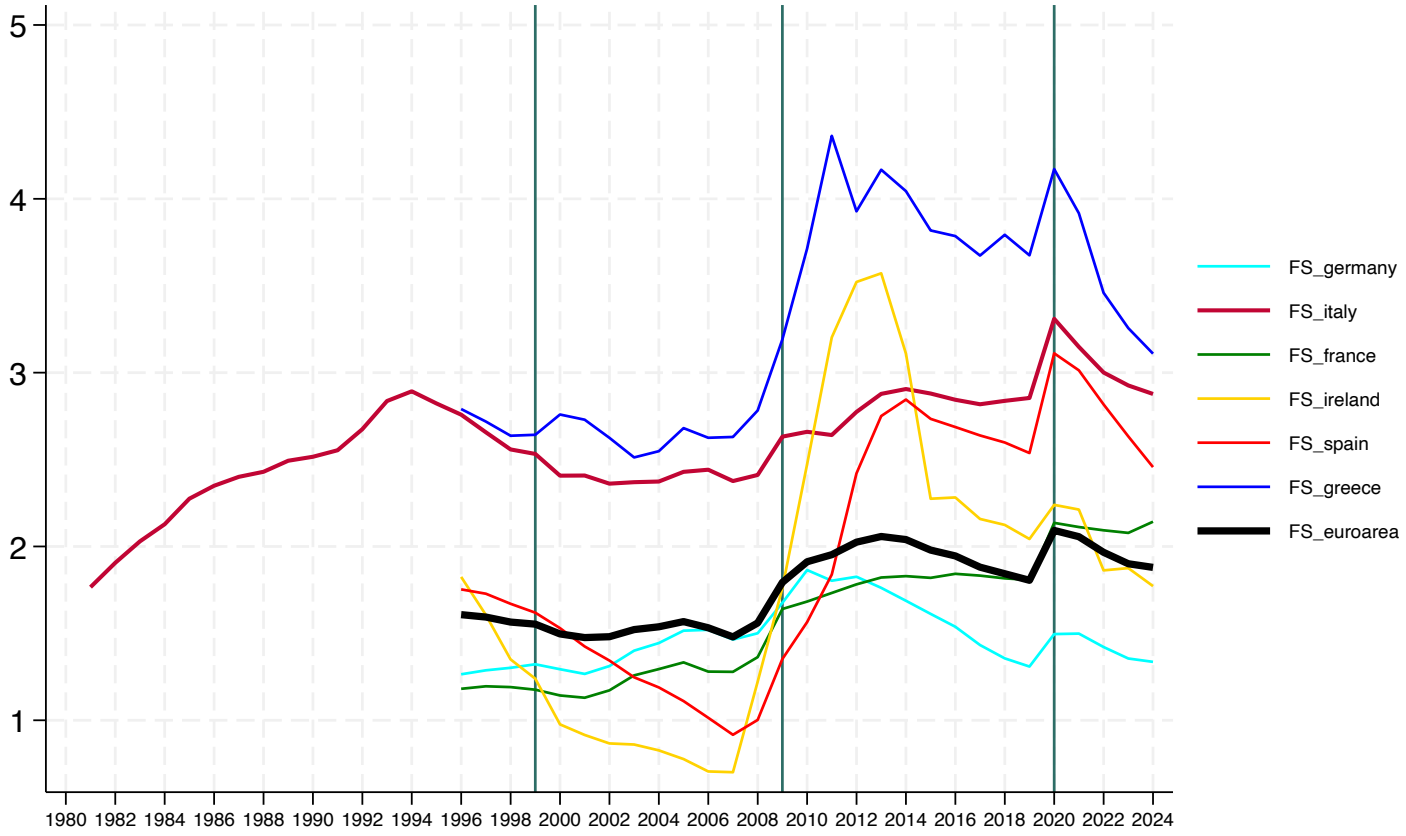
Primary deficit/GDP



Source: Own elaboration on BdI data

De facto fiscal space = $\frac{\text{(public debt)}}{\text{tax base}}$

The fiscal space is weak when debt is high relative to the tax base.
This measure is used by Aizenman et al (2013)

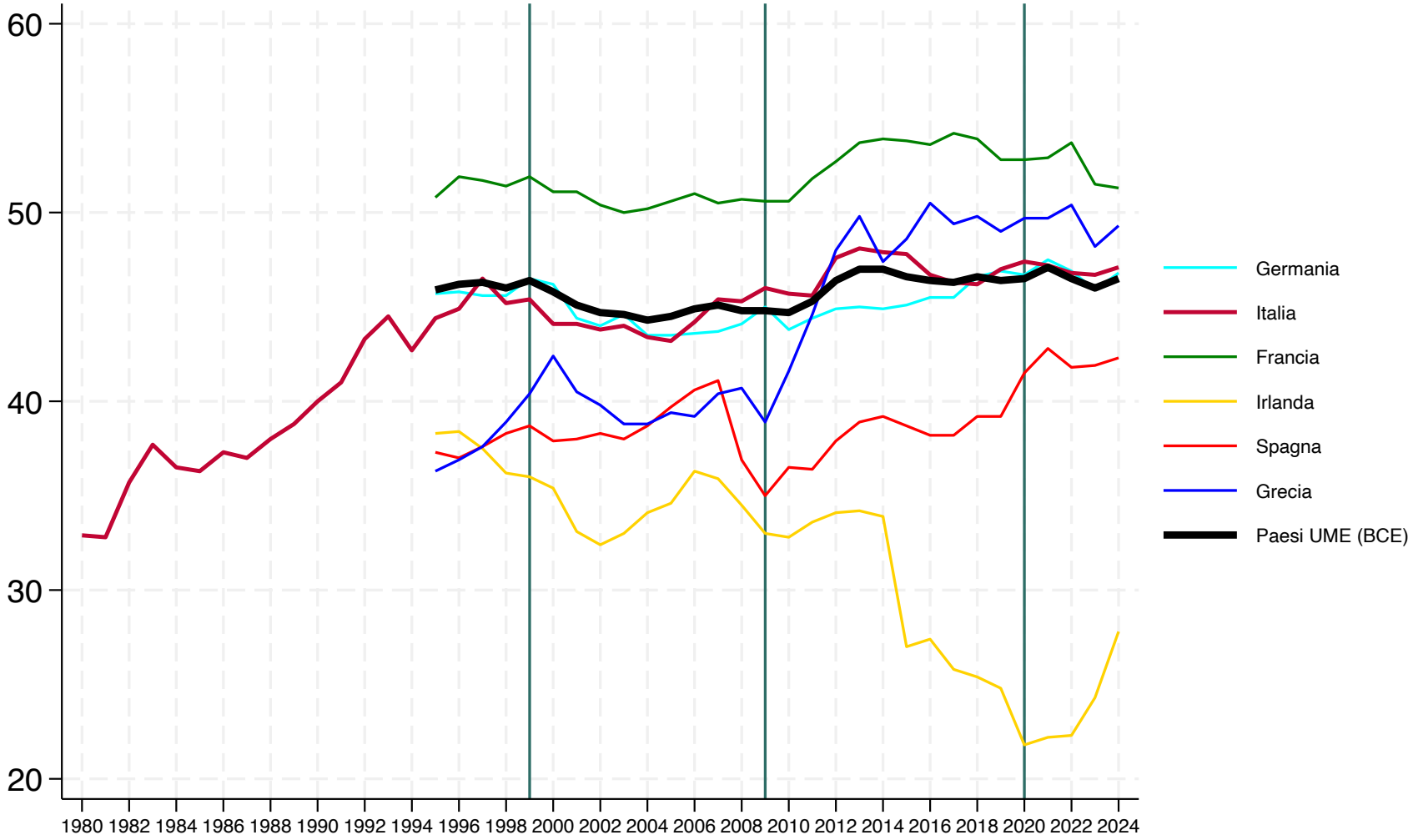


Note: tax base = 5-year moving average of (total revenues/GDP)

Source: Own elaboration on BdI data

The notion of fiscal space has to do with the concept of fiscal sustainability that can be defined as the *"room in a government's budget that allows it to provide resources for a desired purpose without jeopardizing the sustainability of its financial position or the stability of the economy."* (Heller 2005)
There can be different measures of fiscal space; the de facto fiscal space is one of those.

FISCAL PRESSURE = TOTAL REVENUES/GDP (percentage)



Source: Own elaboration on BdI data

Back to Maastricht Treaty: the final phase

Table 6.1 Conversion rates of EMU currencies into the euro

Belgian franc	40.339900
Spanish peseta	166.386000
Irish punt	0.787564
Luxembourgish franc	40.339900
Austrian schilling	13.760300
Finnish marka	5.945730
German mark	1.955830
French franc	6.559570
Italian lire	1936.270000
Dutch guilder	2.203710
Portuguese escudo	200.482000

How to organize relations between the ‘ins’ and the ‘outs’ : main principles

- Main principles decided in ECOFIN meeting in June 1996:
  - A new exchange rate mechanism (the so-called ERM-II) has replaced the old Exchange Rate Mechanism (ERM) since 1 January 1999.
  - Adherence to the mechanism is voluntary.
  - Its operating procedures are determined in agreement between the ECB and the central banks of the 'outs'.
  - ERM-II is based on central rates around which +/- 15% margins of fluctuations are set. Countries may choose different (lower) margins.

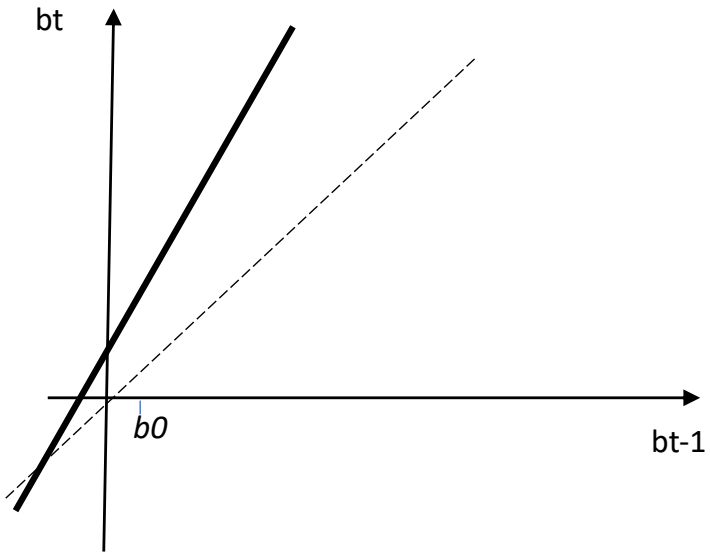
- The anchor of the system is the euro.
- When the exchange rates reach the limit of the fluctuation margin, intervention (i.e. buying or selling) is mandatory.
- This obligation will be dropped if the interventions conflict with the objectives of price stability in the Eurozone or in the outside country.
- The ECB has the power to initiate a procedure aimed at changing the central rates.
- Although participation in ERM II is voluntary, as one of the convergence criteria for entry to the euro area, a country must participate in the mechanism without severe tensions and without devaluing its central rate against the euro on its own initiative for at least two years before it can qualify to adopt the euro.

ERM II today

- Currently, ERM II includes the currencies of Bulgaria and Denmark.
- The Bulgarian lev joined ERM II on 10 July 2020 and observed a central rate of 1.95583 to the euro. Bulgaria then joined the euro on January 1, 2026.
- Following the Croatian kuna joining ERM II on 10 July 2020, Croatia adopted the euro on 1 January 2023.
- The Danish kroner joined ERM II on 1 January 1999, and observes a central rate of 7.46038 to the euro with a narrow fluctuation band of ±2.25%.
- Sweden has not adhered to the ERM-II.

Exercises on debt dynamics

1. Consider the debt/GDP dynamics depicted below, where *b0* is the level of the debt/GDP ratio of the country at the starting year 0. How do you expect the debt/GDP ratio to change in the following two years if the real interest rate and the GDP growth rate remain constant? What could the government do to reduce the debt/GDP ratio?



Let ***r*** be the real interest rate and ***g*** the GDP growth rate. Use the graphical analysis to show that when ***g>r*** the dynamics of the debt/GDP ratio converge to a stable equilibrium.

Use the graphical analysis and/or maths to explain under which conditions a reduction of Deficit/GDP can bring about a reduction of Debt/GDP when r>g